

JAVA PROJECT REPORT ON

CREDIT CARD BILLING AND PAYMENT REMINDER SYSTEM

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DECLARATION

I hereby declare that the project entitled "**CREDIT CARD BILLING AND PAYMENT REMINDER SYSTEM**" submitted in partial fulfillment of the requirements for the course [**INDHU/B.SC.IT**] is a **record of my original work** carried out under the guidance of **[DHILEEBAN A]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

INDHU K

CERTIFICATE

This is to certify that the project entitled **"CREDIT CARD BILLING AND PAYMENT REMINDER SYSTEM"** is a **bona fide work** carried out by **[INDHU] [B.SC.IT/]**, [**Department Name**], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period April 2026.

Place:COIMBATORE

Date:9/4/2026

Submitted for the Viva-Voce Examination held on _____

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The **Credit Card Billing and Payment Reminder System** is a Java-based application designed to automate the management of credit card accounts, billing cycles, and payment reminders. In many financial institutions and individual users, tracking credit card dues, minimum payments, and due dates is done manually or through generic tools, which can lead to missed payments, interest penalties, and poor financial management. This project aims to simplify credit card billing operations by providing an automated system that manages account balances, tracks transactions, calculates interest, and sends timely payment reminders.

The system allows users to add credit card accounts, record transactions, view billing summaries, and receive automated payment reminders based on due dates. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, collections, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface**.

The main objective of the system is to improve billing accuracy, prevent missed payments, reduce interest charges, and provide users with a clear overview of their credit card dues. The system generates detailed billing statements that display outstanding balances, minimum due amounts, payment due dates, and transaction history.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a foundation for developing more advanced personal finance management systems in the future.

1. INTRODUCTION

In the modern financial world, credit cards have become an essential tool for managing everyday expenses. With the growing use of credit cards, keeping track of multiple card balances, transaction histories, billing cycles, and payment due dates has become increasingly challenging for users. Missing a payment deadline can result in heavy interest penalties, late fees, and damage to one's credit score.

With the advancement of computer technology, automated billing and payment reminder systems have become essential for managing credit card obligations efficiently. A computerized system helps users track their outstanding dues, receive timely reminders, and manage their payments without manual intervention.

The Credit Card Billing and Payment Reminder System is a Java-based application designed to automate the process of credit card account management, billing cycle tracking, interest calculation, and payment reminder generation. The system allows users to register their credit card accounts, record debit and credit transactions, view current outstanding balances, and receive automated alerts for upcoming payment due dates.

This project is developed using **Core Java programming concepts** such as classes, objects, methods, conditional statements, loops, ArrayList, and exception handling. The application can be implemented either as a **console-based program or using Java Swing for a graphical user interface (GUI)**.

The primary goal of this system is to reduce missed payments, improve billing accuracy, and provide users with a reliable platform for managing their credit card finances. This project also demonstrates how Java programming can be applied to develop practical financial management applications.

1.1 Objective

The main objective of the **Credit Card Billing and Payment Reminder System** is to develop a simple and efficient billing application using Core Java that automates the process of credit card account management, billing cycle handling, and payment reminder generation.

The specific objectives of this project are:

- To design and develop a Java-based credit card billing management system.
- To automate the calculation of outstanding balances, minimum payments, and interest charges.
- To generate payment reminders based on billing due dates.
- To provide a user-friendly interface for managing multiple credit card accounts.
- To demonstrate the implementation of Core Java programming concepts such as classes, objects, methods, loops, collections, and conditional statements.
- To generate a clear and organized billing statement for each credit card account.

1.2 Overview

The Credit Card Billing and Payment Reminder System is a menu-driven application developed using the Java programming language. The system provides a digital platform where users can manage their credit card accounts, track transactions, and monitor payment due dates.

The system displays a list of registered credit card accounts along with their outstanding balances and due dates. Users can record new transactions, view current billing summaries, and receive automated reminders when payment due dates are approaching. The system calculates the minimum due amount and any applicable interest charges based on the outstanding balance.

The system helps simplify financial management by automating billing calculations and sending timely payment alerts. It can be developed as a console-based program or using Java Swing for a graphical user interface.

This project demonstrates the practical use of Java programming in developing real-world financial management applications.

1.3 Features

The Credit Card Billing and Payment Reminder System includes the following main features:

- **Account Management – Allows users to add, view, and manage multiple credit card accounts with individual credit limits and billing cycles.**
- **Transaction Recording – Enables users to record purchases, payments, and refunds for each credit card account.**
- **Automatic Balance Calculation – Automatically calculates the outstanding balance after each transaction.**
- **Interest Calculation – Calculates monthly interest charges based on the outstanding balance and applicable interest rate.**
- **Billing Statement Generation – Generates a detailed billing statement showing transactions, outstanding balance, minimum due, and payment due date.**
- **Payment Reminder – Sends automated reminders when the payment due date is approaching.**
- **User-Friendly Interface – Provides a simple and easy-to-use menu-driven interface for managing credit card finances.**

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Many credit card holders struggle to keep track of their dues, billing cycles, and payment deadlines across multiple cards. Without a dedicated management system, users often rely on manual tracking through spreadsheets or paper records, which is inefficient and prone to errors.

Some common problems include:

- Difficulty tracking multiple credit card balances and due dates
- Missed payment deadlines resulting in late fees and interest charges
- Lack of automated reminders for upcoming payment due dates
- Manual calculation errors in interest and minimum payment computation
- Absence of a consolidated billing statement for each account

Therefore, there is a need for an automated system that can manage credit card accounts, calculate balances and interest, and send timely payment reminders. The **Credit Card Billing and Payment Reminder System** is designed to address these issues.

2.2 Project Overview

The Credit Card Billing and Payment Reminder System is a Java-based application developed to simplify credit card billing management for individual users and small financial institutions.

The system provides a menu-driven interface where users can manage their credit card accounts, record transactions, view billing statements, and receive payment reminders. The system automatically calculates the outstanding balance, minimum payment due, and applicable interest charges.

The system includes the following functionalities:

- Adding and managing credit card accounts
- Recording transactions (purchases, payments, refunds)
- Calculating outstanding balance and interest charges
- Generating billing statements
- Sending payment due date reminders

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The system can be used in:

- Personal finance management by individual credit card users
- Small financial institutions managing customer billing records
- Educational institutions for demonstrating Java programming applications
- Banking and fintech training programs

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Credit Card Billing and Payment Reminder System is designed as a menu-driven application where users can manage card accounts, record transactions, view billing statements, and receive payment reminders automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Credit Card Billing and Payment Reminder System** are explained below:

1. Account Management Module

This module manages the registration and storage of credit card account details.

Functions of this module include:

- Adding new credit card accounts with card number, holder name, credit limit, and interest rate
- Viewing existing account details
- Updating account information

2. Transaction Management Module

This module handles recording and tracking of all transactions for each credit card account.

Functions of this module include:

- Recording purchase transactions
- Recording payment transactions
- Recording refund transactions
- Displaying transaction history

3. Billing Calculation Module

This module performs all billing calculations for each credit card account.

The system calculates:

- Outstanding balance after each transaction
- Monthly interest charges based on the balance and interest rate
- Minimum payment due amount
- Total amount payable including interest

Example Calculation:

```
Outstanding Balance = Previous Balance + Purchases - Payments
Monthly Interest = Outstanding Balance × (Annual Rate / 12)
Minimum Due = 5% of Outstanding Balance or ₹500, whichever is higher
```

4. Billing Statement Module

This module generates the detailed billing statement for each credit card account.

The statement includes:

- Cardholder name and card number
- Statement period
- List of transactions with dates and amounts
- Outstanding balance
- Interest charges
- Minimum payment due
- Payment due date

Example Statement Output:

```
------ CREDIT CARD BILLING STATEMENT ------
Cardholder: Arun Kumar       Card: XXXX-XXXX-XXXX-4521
Statement Period: 01-Mar-2026 to 31-Mar-2026

Date         Description          Amount
10-Mar-2026  Online Shopping       ₹4,500
18-Mar-2026  Restaurant             ₹1,200
25-Mar-2026  Payment Received     -₹3,000

Outstanding Balance:          ₹2,700
Interest (3.5% p.m.):          ₹94.50
Minimum Due:                   ₹500
Payment Due Date:         15-Apr-2026
```

5. Payment Reminder Module

This module checks the payment due dates of all registered credit card accounts and generates reminders when the due date is approaching.

The reminders are triggered:

- 7 days before the payment due date
- 3 days before the payment due date
- On the payment due date

6. Exit Module

This module allows the user to exit the application safely after completing the required operations.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Credit Card Billing and Payment Reminder System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of credit card users and financial management operations.

The main problems identified in manual credit card management are:

- Manual tracking of multiple card balances leads to errors and missed payments.
- No automated reminders for upcoming due dates.
- Difficulty in calculating interest and minimum payment amounts accurately.
- Lack of consolidated billing statements for each account.

Based on these problems, system requirements were defined to develop an automated credit card billing and reminder application.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Account Management Module
- Transaction Management Module
- Billing Calculation Module
- Billing Statement Module
- Payment Reminder Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

- Creation of CreditCard and Transaction classes
- Implementation of menu-driven account management logic
- Writing code for billing calculations and interest computation
- Developing the payment reminder logic using date comparison
- Generating billing statements in formatted output

Core Java concepts such as **classes, objects, ArrayList, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Add credit card account	Account saved and displayed correctly
Record a purchase	Balance updated correctly
Record a payment	Balance reduced correctly
Calculate interest	Correct interest amount computed
Generate billing statement	Correct statement displayed
Payment reminder check	Reminder triggered on due date
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates accurate billing statements and payment reminders based on the credit card accounts and transaction data entered.

The results are demonstrated through program execution and screenshots showing the input and output of the system. Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Credit Card Billing and Payment Reminder System**:

1. Start the program.
2. Display the main menu with options: Add Account, Record Transaction, View Statement, Check Reminders, Exit.
3. If the user selects 'Add Account', enter card details and store the account.
4. If the user selects 'Record Transaction', select the account and enter transaction details.

5. Update the outstanding balance after each transaction.
6. If the user selects 'View Statement', calculate interest and minimum due, then display the billing statement.
7. If the user selects 'Check Reminders', compare today's date with payment due dates.
8. If the due date is within 7 days, display a payment reminder.
9. Repeat the menu until the user selects 'Exit'.
10. End the program.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Display Main Menu
↓
Select Option
↓
Perform Operation (Add Account / Record Transaction / View Statement / Check Reminder)
↓
Display Output
↓
Exit?
↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import java.util.*;

class CreditCard {
   String cardHolderName;
   String cardNumber;
   double dueAmount;
   String dueDate;


   CreditCard(String name, String number, double amount, String date) {
      cardHolderName = name;
      cardNumber = number;
      dueAmount = amount;
      dueDate = date;
   }


   void displayDetails() {
      System.out.println("Card Holder Name: " + cardHolderName);
      System.out.println("Card Number: " + cardNumber);
      System.out.println("Due Amount: ₹" + dueAmount);
      System.out.println("Due Date: " + dueDate);
   }


   void sendReminder() {
      System.out.println("Reminder: Dear " + cardHolderName +
            ", your payment of ₹" + dueAmount +
            " is due on " + dueDate + ". Please pay on time!");
   }
}

public class CreditCardReminderSystem {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter Card Holder Name: ");
      String name = sc.nextLine();

      System.out.print("Enter Card Number: ");
      String number = sc.nextLine();

      System.out.print("Enter Due Amount: ");
      double amount = sc.nextDouble();
      sc.nextLine(); /

      System.out.print("Enter Due Date (DD-MM-YYYY): ");
      String date = sc.nextLine();


      CreditCard card = new CreditCard(name, number, amount, date);

      System.out.println("\n--- Credit Card Details ---");
```

```
        card.displayDetails();

        System.out.println("\n--- Payment Reminder ---");
        card.sendReminder();

        sc.close();
    }
}
```

6. INPUT AND OUTPUT SCREENS

INPUT

```
Enter Card Holder Name: Indhu
Enter Card Number: 1234 5678 9012 3456
Enter Due Amount: 5000
Enter Due Date (DD-MM-YYYY): 25-04-2026
```

OUTPUT

```
--- Credit Card Details ---
Card Holder Name: Indhu
Card Number: 1234 5678 9012 3456
Due Amount: ₹5000.0
Due Date: 25-04-2026

--- Payment Reminder ---
Reminder: Dear Indhu, your payment of
₹5000.0 is due on 25-04-2026. Please pay
on time!
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Credit Card Billing and Payment Reminder System was developed to automate the management of credit card accounts, billing cycles, and payment reminders. The system allows users to add card accounts, record transactions, view billing statements, and receive automated payment reminders.

During the development process, Core Java programming concepts such as classes, objects, ArrayList, loops, conditional statements, and date handling were applied. The system successfully performs billing calculations, generates accurate statements, and provides timely payment reminders.

The system was tested with different inputs to ensure that it performs the required calculations correctly. The results show that the system reduces manual tracking effort, prevents missed payments, and improves overall financial management efficiency.

7.2 Conclusion

The Credit Card Billing and Payment Reminder System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world financial management application.

The system automates credit card billing, calculates interest charges, generates billing statements, and provides timely payment reminders. It simplifies credit card management by giving users a clear view of their dues and helping them avoid late payment penalties.

This project helps students understand how Java programming can be used to develop practical software systems for financial management. The project can be further enhanced by adding features such as database connectivity for persistent storage, graphical user interfaces using Java Swing, email or SMS notification for reminders, and integration with online banking APIs.

8. REFERENCES

Books

- Herbert Schildt – Java: The Complete Reference
- Cay S. Horstmann – Core Java Volume I: Fundamentals

Websites

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